

AI

AI Security Defense Dashboard

Network Security & AI Defense

Group-2:

1. Suon Pisey
2. Nem Sothea
3. Sourn Savourn
4. Oun Sreynich
5. Moeun Nithvaraman

TEST CASES COVERED

BF

Brute Force Login

PS

Port Scan

DD

DDoS Flood

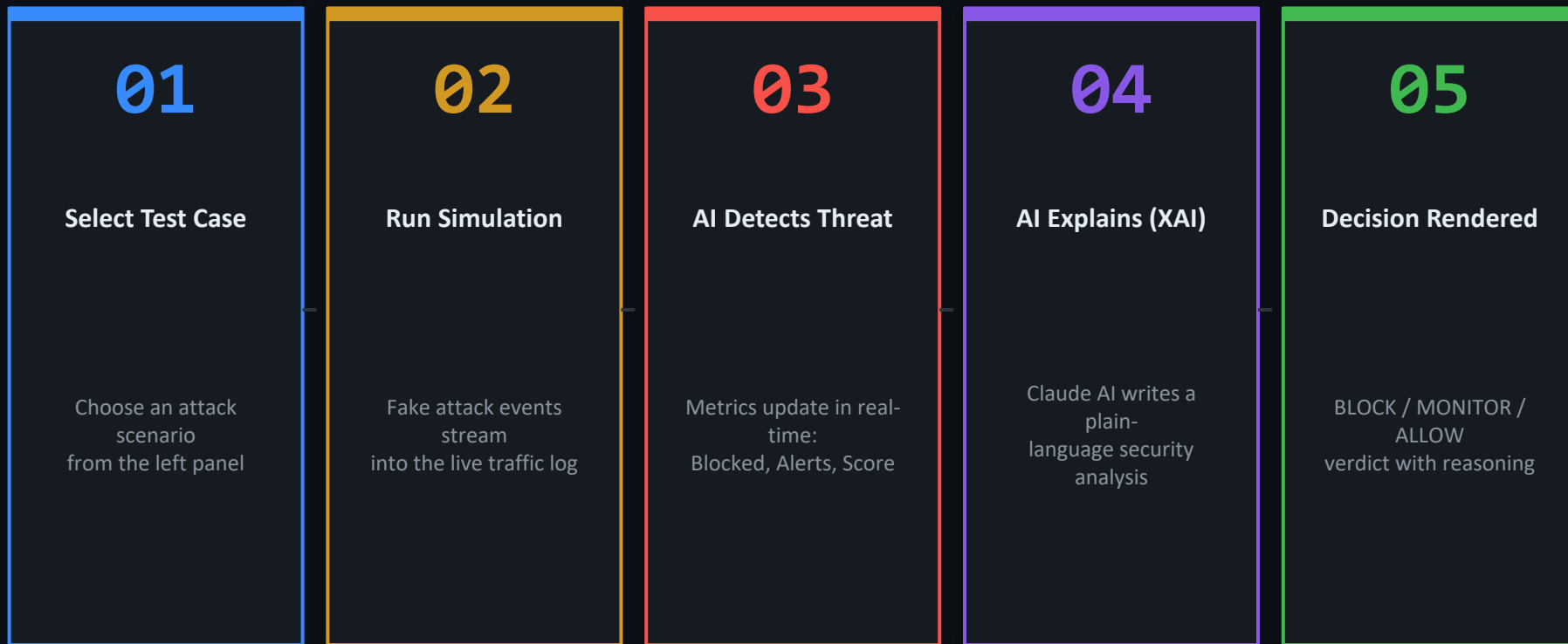
SQ

SQL Injection

AD

Adversarial Attack

How the Dashboard Works



All attacks are 100% simulated – but real world cases. Safe to run in any environment.

BF

Brute Force Login

TEST CASE 01

WHAT IS IT?

An attacker tries thousands of username/password combinations rapidly until they find the correct one. Automated bots can attempt hundreds of logins per second.

HOW AI DETECTS IT

- Login rate exceeds 4 attempts/sec
- Sequential username enumeration
- Same IP, high entropy passwords
- Geolocation anomaly detected

LIVE LOG SIMULATION

14:22:01	103.45.12.88	Failed login – user: admin (1/5)
14:22:02	103.45.12.88	Failed login – user: root (2/5)
14:22:03	103.45.12.88	Failed login – user: administrator (3/5)
14:22:03	AI Engine	Analyzing login frequency...
14:22:04	103.45.12.88	Login from new geolocation: CN
14:22:05	Firewall	Rate limit exceeded – IP blocked
14:22:05	WAF	Credential stuffing blocked
14:22:06	Auth	Account locked – 5 failed attempts



AI DECISION: BLOCK — Threat Confirmed

PS

Port Scan

TEST CASE 02

WHAT IS IT?

An attacker scans all 65,535 ports on a target machine to discover which services are running. This is reconnaissance — mapping the attack surface before launching a real exploit.

HOW AI DETECTS IT

- 1,240 SYN packets in 3 seconds
- No completed TCP handshakes
- Sequential port ordering pattern
- Known Nmap OS fingerprint headers

LIVE LOG SIMULATION

09:15:01	185.220.101.45	SYN → port 22 (SSH)
09:15:01	185.220.101.45	SYN → port 80 (HTTP)
09:15:01	185.220.101.45	SYN → port 443 (HTTPS)
09:15:02	IDS	Rapid sequential port probe
09:15:02	185.220.101.45	SYN → port 3306 (MySQL)
09:15:02	185.220.101.45	SYN → port 6379 (Redis)
09:15:03	IDS	SYN flood – 1,240 pkts in 3s
09:15:03	Firewall	Stealth scan blocked – IP blocked



AI DECISION: BLOCK — Reconnaissance Detected

DD

DDoS Flood

TEST CASE 03

WHAT IS IT?

Distributed Denial of Service: thousands of compromised machines (a botnet) flood your server with traffic, consuming all bandwidth and making the service unavailable to legitimate users.

HOW AI DETECTS IT

- 8,200 req/s (68x normal baseline)
- 1,400+ unique source IPs
- Mixed UDP / SYN / ICMP vectors
- Botnet C2 signature matched

8,200

req/sec

1,400+

source IPs

1.4%

legit traffic

LIVE LOG SIMULATION

16:44:01	Multiple	Traffic spike: 8,200 req/s
16:44:01	Sensor	UDP flood from 1,400+ IPs
16:44:02	Firewall	SYN flood mitigated at edge
16:44:02	BGP	Blackhole route engaged
16:44:03	DNS	Amplification vector neutralized
16:44:03	Firewall	2,100 botnet IPs rate-limited
16:44:04	CDN	Scrubbing center activated
16:44:05	Gateway	Legitimate traffic passed (1.4%)

**AI DECISION: BLOCK — Upstream Mitigation**

WHAT IS IT?

Malicious SQL code is inserted into input fields to manipulate the database. Attackers can extract all data, delete tables, or bypass authentication entirely.

HOW AI DETECTS IT

- UNION SELECT in GET parameters
- Time-based blind injection probes
- sqlmap user-agent fingerprint
- 34 distinct injection vectors found

EXAMPLE PAYLOADS

```
GET /search?q=1' OR '1'='1'--  
' ; DROP TABLE users; --
```

LIVE LOG SIMULATION

11:30:01	91.234.99.12	GET /search?q=1' OR '1'='1'--
11:30:01	WAF	SQL injection blocked: UNION SELECT
11:30:02	91.234.99.12	Time-based blind SQLi detected
11:30:02	WAF	Error-based SQLi neutralized
11:30:03	91.234.99.12	Payload: ' ; DROP TABLE users;--
11:30:03	WAF	sqlmap user-agent matched
11:30:04	91.234.99.12	Database error disclosure attempt
11:30:04	WAF	Stacked query injection blocked



AI DECISION: BLOCK — Database Attack Confirmed

AD

Adversarial Attack

TEST CASE 05

WHAT IS IT?

An advanced attacker crafts inputs specifically designed to fool the AI detection system itself — probing its decision boundaries and using encoded payloads to bypass ML-based security.

HOW AI DETECTS IT

- Unusual inter-packet timing patterns
- Entropy anomalies in HTTP body
- Multi-layer encoding (Base64+ROT13)
- GAN-generated traffic signatures

WHY IT'S UNIQUE

The attacker targets the AI itself — not the system. This is why Explainable AI (XAI) and adversarial training matter.

LIVE LOG SIMULATION

20:05:01	172.33.45.11	Evasion: fragmented packet stream
20:05:02	172.33.45.11	AI model probing with crafted inputs
20:05:02	ML Engine	Adversarial payload fingerprinted
20:05:03	172.33.45.11	Obfuscated shellcode in HTTP body
20:05:03	WAF	Base64+ROT13 payload blocked
20:05:04	172.33.45.11	Model poisoning: /api/predict
20:05:04	IDS	Header manipulation blocked
20:05:05	Sandbox	Sample sent for adversarial training

◆ AI DECISION: MONITOR — Feed to Retraining Pipeline

Explainable AI (XAI)

How the AI explains its decisions

What AI Sees

- Network traffic patterns
- Packet timing & entropy
- Source IP reputation
- Payload signatures
- Login frequency rate

How AI Decides

- Compares against trained models
- Scores anomaly level 0-100%
- Checks threat intelligence feeds
- Calculates confidence score
- Applies behavioral rules

What AI Reports

- Plain-language explanation
- Key indicators triggered
- Threat & confidence scores
- BLOCK / MONITOR / ALLOW
- Recommended next action

Powered by Claude AI (Anthropic API) – the AI reads simulated attack data and writes a real-time security analysis in plain language for each test case.

Key Takeaways

AI Security Defense
Dashboard – Goup-2

Network Security & AI Defense

01

AI detects threats faster than humans by analyzing thousands of packets per second automatically.

02

All 5 attack types (Brute Force, Port Scan, DDoS, SQLi, Adversarial) require different detection strategies.

03

Explainable AI (XAI) is critical — security teams must understand WHY the AI made a decision.

04

AI systems themselves must be protected. Adversarial attacks specifically target the AI model.

05

Organizations must combine AI detection with human oversight for reliable security operations.